

Genetic Analysis Report

Report ID:	SEQ-2B1A8F56D208
Generated:	2025-12-04 14:46:51
Patient Name:	Genetic Risk Analysis Report - HIGH RISK

Analysis Accuracy

94.2%

Patient Information

Analysis performed on file DNA-5ED904266D81. Processing completed using GENO AI v2.0 Elastic Net Logistic Regression model. Model trained on gene expression data with 94.2% cross-validation accuracy. Risk probability calculated: 95.0%

Variant Information

Gene Expression Analysis Complete • Analyzed Genes: 237 • Mean Expression Level: 52.99 • Expression Variance: 57661.55 • High Expression Markers: 59 • Risk Assessment: HIGH RISK • Confidence: 95.0%

Analysis Results

{"report": {"sample_id": "SEQ-2B1A8F56D208", "laboratory_user_id": "F55055", "generated_at": "2025-12-04T11:46:51.448182Z", "model_name": "geno_enet_pipeline.pkl", "model_version": "2.0", "total_genes_in_model": 237, "genes_analyzed": 237, "risk_level": "LOW", "risk_score_percent": 68.64, "risk_score_raw": 0.6864262306551446, "risk_threshold": 75.0, "summary_title": "Genetic Risk Analysis Report - LOW RISK", "summary_text": "The genetic analysis indicates a LOW addiction risk profile based on the evaluated gene-expression markers. The model analyzed 237 genetic markers and calculated a risk probability of 68.64%. This score is below the 75% threshold, indicating lower risk.", "top_genes": [{"gene": "ENSG00000140264.21", "expression": 2687.0}, {"gene": "ENSG00000125356.7", "expression": 1386.0}, {"gene": "ENSG00000227544.10", "expression": 1169.0}, {"gene": "ENSG00000159788.20", "expression": 1049.0}, {"gene": "ENSG00000164604.14", "expression": 798.0}, {"gene": "ENSG00000100216.6", "expression": 711.0}, {"gene": "ENSG00000110092.4", "expression": 603.0}, {"gene": "ENSG00000099795.7", "expression": 578.0}, {"gene": "ENSG00000232388.5", "expression": 571.0}, {"gene": "ENSG00000150779.12", "expression": 452.0}], "bottom_genes": [{"gene": "ENSG00000280027.1", "expression": 0.0}, {"gene": "ENSG00000196893.3", "expression": 0.0}, {"gene": "ENSG00000287302.1", "expression": 0.0}, {"gene": "ENSG00000283145.1", "expression": 0.0}, {"gene": "ENSG00000243029.3", "expression": 0.0}], "gene_stats": {"mean_expression": 52.9916, "median_expression": 0.0, "std_expression": 240.6364, "min_expression": 0.0, "max_expression": 2687.0, "genes_above_mean": 26,

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This report was generated automatically from genetic data analysis. For questions or concerns, please contact your healthcare provider.